Determinants

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Determinants assign a signed volume and test invertibility. Their defining properties yield row-operation rules, cofactor expansions, permutation formulas, and inverse formulas.

Definition and Properties

$$\det I = 1, \qquad \det(AB) = \det(A)\det(B), \qquad A \text{ invertible} \iff \det A \neq 0.$$

Row rules:
swapping rows changes the sign; multiplying a row by c multiplies the determinant by c; adding a multiple of one row to another leaves it unchanged.

Linearity:
the determinant is linear in each row (or column) separately and alternating in its rows.

$$\det(A^T) = \det A, \qquad \det(A^{-1}) = (\det A)^{-1}, \qquad \det(cA) = c^n \det A.$$

Computational Formulae

Triangular matrices:
$\det A = \prod_{i=1}^n a_{ii}$.

$$\det A = \sum_{j=1}^n a_{ij} C_{ij} = \sum_{i=1}^n a_{ij} C_{ij}, \quad C_{ij} = (-1)^{i+j} \det A_{ij}.$$

Cofactor formula:
$A^{-1} = \dfrac{1}{\det A} \operatorname{Cof}(A)^T$.

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}.$$

Cramer's rule:
if $Ax = b$ and $\det A \neq 0$, $x_i = \dfrac{\det A_i(b)}{\det A}$.

Vandermonde determinant:
$\det[x_j^{i-1}]_{i,j=1}^n = \displaystyle\prod_{1 \le i < j \le n} (x_j - x_i)$.

Geometry and Volume

$$\operatorname{volume}(A[0,1]^n) = |\det A|, \qquad \det A = 0 \iff \text{columns lie in a lower-dimensional subspace.}$$